

Redlich–Kwong equation of state

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The Redlich–Kwong equation of state is a two-constant equation:

$$p = \frac{RT}{V_m - b} - \frac{a}{T^{1/2} V_m (V_m + b)}$$

- p is pressure
- T is the absolute temperature
- R is the gas constant
- V_m is the molar volume
- a is a constant reflecting attraction between molecules
- b is the excluded volume of molecules

Constants a , b depend on the critical temperature and pressure via equations:

$$a = \frac{1}{9(2^{1/3} - 1)} \frac{R^2 T_c^{5/2}}{p_c}, \quad b = \frac{2^{1/3} - 1}{3} \frac{RT_c}{p_c}$$

- Actually, a , b are back-calculated from the experimental p_c , T_c to use the equation!

- accuracy ~ 2 to 5%
- simple
- the critical compressibility factor

$$z_c = \frac{p_c V_c}{RT_c} = \frac{1}{3}$$

is much better than for the van der Waals equation ($z_c = 3/8$)

for most gases (excl. water vapor): $z_c \approx 0.28\text{--}0.30$

- bad near the critical point: no classical mean-field equation of state can handle it

The Redlich–Kwong equation is cubic in volume. Thus, it has up to three roots:

- one real root for $T > T_c$
- a real triple root for $T = T_c$ (singular case)
- up to three real roots for $T < T_c$:
 - liquid (stable or metastable)
 - vapor (stable or metastable)
 - unstable phase in between (immediate spinodal decomposition)
 - if there are 2 or more roots, the stable root has the lowest fugacity
- The JavaScript applet calculator uses the secant method to find the roots.

- The fugacity coefficient

$$\ln \phi = \int_{V_m(p)}^{\infty} \left[\frac{p(V_m)}{RT} - \frac{1}{V_m} \right] dV_m - \ln z + z - 1 = \int_0^{c(p)} \left[\frac{p(1/c)}{RT} - c \right] \frac{dc}{c^2} - \ln z + z - 1$$

Substitution $c = 1/V_m$ is suitable for numerical integration.

- For Redlich–Kwong, the integration can be performed analytically:

$$p = p(V_m, T) = \frac{RT}{V_m - b} - \frac{a}{\sqrt{T} V_m (V_m + b)}$$

$$z = \frac{V_m}{V_m - b} - \frac{a}{RT^{3/2} (V_m + b)},$$

$$\ln \phi = \ln \frac{V_m}{V_m - b} + \frac{a}{RT^{3/2} b} \ln \frac{V_m}{V_m + b} - \ln z + z - 1$$

- The fugacity $f = \phi p$ is used to determine stability for $T < T_c$